TOPIC 8.1
Electric Charge and Electric Force

Required Course Content

LEARNING OBJECTIVE

8.1.A
Describe the electric force that results from the interactions between charged objects or systems.

ESSENTIAL KNOWLEDGE

8.1.A.1
Charge is a fundamental property of all matter.

8.1.A.1.i
Charge is a scalar quantity and is described as positive or negative.

8.1.A.1.ii
The magnitude of the charge of a single electron or proton, the elementary charge e, can be considered to be the smallest indivisible amount of charge.

8.1.A.1.iii
The charge of an electron is $-e$ and the charge of a proton is $+e$, and a neutron has no electric charge.

8.1.A.1.iv
A point charge is a model in which the physical size of a charged object or system is negligible in the context of the situation being analyzed.

8.1.A.2
Coulomb's law describes the electrostatic force between two charged objects as directly proportional to the magnitude of each of the charges and inversely proportional to the square of the distance between the objects.

Relevant equation:

$$\left|\vec{F}_E\right| = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

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SUGGESTED SKILLS

1.A
Create diagrams, tables, charts, or schematics to represent physical situations.

2.B
Calculate or estimate an unknown quantity with units from known quantities, by selecting and following a logical computational pathway.

2.D
Predict new values or factors of change of physical quantities using functional dependence between variables.

3.B
Apply an appropriate law, definition, theoretical relationship, or model to make a claim.

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Electric Charges, Fields, and Gauss's Law

LEARNING OBJECTIVE	ESSENTIAL KNOWLEDGE
8.1.A Describe the electric force that results from the interactions between charged objects or systems.	**8.1.A.3** The direction of the electrostatic force depends on the signs of the charges of the interacting objects and is along the line of separation between the objects. **8.1.A.3.i** Two objects with charges of the same sign exert repulsive forces on each other. **8.1.A.3.ii** Two objects with charges of opposite signs exert attractive forces on each other. **8.1.A.4** Electric forces are responsible for some of the macroscopic properties of objects in everyday experiences. However, the large number of particle interactions that occur make it more convenient to treat everyday forces in terms of nonfundamental forces called contact forces, such as normal force, friction, and tension.
8.1.B Describe the electric and gravitational forces that result from interactions between charged objects with mass.	**8.1.B.1** Electrostatic forces can be attractive or repulsive, while gravitational forces are always attractive. **8.1.B.2** For any two objects that have mass and electric charge, the magnitude of the gravitational force is usually much smaller than the magnitude of the electrostatic force. **8.1.B.3** Gravitational forces dominate at larger scales even though they are weaker than electrostatic forces, because systems at large scales tend to be electrically neutral.
8.1.C Describe the electric permittivity of a material or medium.	**8.1.C.1** Electric permittivity is a measurement of the degree to which a material or medium is polarized in the presence of an electric field. **8.1.C.2** Electric polarization can be modeled as the induced rearrangement of electrons by an external electric field, resulting in a separation of positive and negative charges within a material or medium. **8.1.C.3** Free space has a constant value of electric permittivity, ε_0, that appears in physical relationships.

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LEARNING OBJECTIVE

8.1.C

Describe the electric permittivity of a material or medium.

ESSENTIAL KNOWLEDGE

8.1.C.4

The permittivity of matter has a value different from that of free space that arises from the matter's composition and arrangement.

8.1.C.4.i

In a given material, electric permittivity is determined by the ease with which electrons can change configurations within the material.

8.1.C.4.ii

Conductors are made from electrically conducting materials in which charge carriers move easily; insulators are made from electrically nonconducting materials in which charge carriers cannot move easily.

BOUNDARY STATEMENT

AP Physics C: Electricity & Magnetism only expects students to make calculations of the electric force between four or fewer interacting charged objects or systems. The analysis of the resulting electric force from more charges is allowed in situations of high symmetry. Note that students are expected to calculate the electric fields of charge distributions, as described in Topics 8.4 and 8.6.

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